

# Tamás Karácsony

PH.D. CANDIDATE

☎ (+351) 913-579-117 | ✉ karacsonytamas.mail@gmail.com | 🏠 tamaskaracsony.github.io/ | 📄 tamas-karacsony | 🎓 Tamás Karácsony

## Summary

I research AI-based diagnosis support systems, focusing on human motion analysis and video-based epileptic seizure classification. My work addresses challenges in action recognition on scarce medical data, requiring efficient data utilization, scalable model design, transfer learning, and synthetic data generation. I integrate AI, biomedical engineering, and neuroscience, applying deep learning to TB-scale clinical datasets while ensuring compliance with medical AI regulations (GDPR, ethics). Over six years of clinical collaboration, my work has led to high-impact publications, two provisional patents, and AI models tested in clinical applications. I am interested in extending action recognition and medical AI models with LLMs and developing generalizable decision support systems, to enhance explainability and adaptability.

**Computer Vision** Action recognition, Human Pose Estimation (HPE), Synthetic Data Augmentation [1]–[7], [9], [11], [12]

**Machine Learning** Vision and Motion Transformers, 3D CNNs, GCNs, LSTM, Transfer learning [1], [3], [4], [6], [8]–[12]

**AI in Healthcare** Diagnosis support, Advanced Human Sensing, Human Motion Analysis [1], [3], [4], [6], [9]–[12]

**Biomedical Engineering** Research and diagnosis support systems for Epilepsy, Neuroengineering, Brain-computer interfaces (BCI) [1]–[12]

## Research Experience

### INESC TEC - Institute for Systems and Computer Engineering, Technology and Science

Porto, Portugal

RESEARCHER AND PH.D. CANDIDATE

Sept. 2019 - Present

- **Machine Learning research:** Developed and trained architectures of action recognition approaches for semiology-based epileptic seizure classification with transfer learning, first based on 3D CNN and LSTM architectures [4], [9], then with 2 stage, Human Pose Estimation (ViTs) [11], [6] and skeleton based action recognition [1], [3], [12] in a distributed computing environment, on GPU clusters. Addressed the blanket occlusion of in-bed HPE with synthetic blanket occlusion data augmentation of HPE datasets for HPE architecture training [11], [6]. Published the research outcomes in leading venues i.e Scientific Reports, Image and Vision Computing.
- **Clinical Collaborations & Data Acquisition:** Research collaboration with clinicians from Neurology departments of Ludwig Maximilian University of Munich (LMU), Germany and São João University Hospital, Porto, Portugal (CHUSJ) on video-based seizure analysis, including setting research agenda for AI-based diagnosis support systems, MLOps and data collection [1], [4], [5], [7]–[9], [12]. Additionally, managed and facilitated the acquisition of multiple novel, large-scale datasets, including 1.2 TB simulated epileptic action dataset [1], [5], 15+ TB clinical 4K RGB-D-IR data, and 720p RGB clinical seizure video data [12]. Developed software for data acquisition systems [5], custom data labeling, processing, and transfer systems [8], and deployed them on-site, while ensuring compliance with GDPR & ethics regulations.
- **Research Leadership & Supervision:** Responsible for setting and carrying out the research agenda of the research line of video-based epileptic seizure diagnosis support. Supervised and co-supervised over the years two contracted researchers, four M.Sc. theses, four B.Sc. internships, and a visiting research scholar in topics of synthetic data generation, and human pose estimation for in-bed scenarios, action and gesture recognition applied for epileptic seizure classification, human motion analysis, MLOps and data acquisition from clinics.
- **Computational Infrastructure:** Managed GPU clusters, Docker containers, and compute resources and advised on hardware requirements based on research needs, code contributions were hosted on institutionally-managed GitLab and internal servers due to regulatory constraints
- **Intellectual Property & Patents:** Two provisional patents in progress related to seizure classification, and AI-based diagnosis support.

RESEARCH ASSISTANT

Febr. - July 2019

- Writing my master thesis: “Motion-based epileptic seizure classification with deep learning”.

## Education

### Carnegie Mellon University

Pittsburgh, USA

VISITING RESEARCH SCHOLAR - CARNEGIE MELLON PORTUGAL AFFILIATED PH.D. PROGRAM

2023 - 2024

- **Advisors:** Prof. Fernando De La Torre and Prof. László A. Jeni
- Worked with Human Pose Estimation (VitPose, MotionBERT), Skeleton and Video based Action recognition (PoseC3D, ST-GCN++, Uniformerv2, I3D) for clinical applications. [1], [3]

### Faculty of Engineering, University of Porto

Porto, Portugal

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING - CARNEGIE MELLON PORTUGAL AFFILIATED PH.D. PROGRAM

2021 - (Expected: September 2025)

- **Advisors:** Prof. João Paulo Cunha (FEUP, INESC TEC) and Prof. Fernando De La Torre (CMU)
- Ph.D. Thesis working title: “Advancing AI in Healthcare: Action Recognition and Human Pose Estimation for Epileptic Seizure Classification” [1], [3]–[9], [11], [12]

## Technical University of Denmark

M.SC. IN BIOMEDICAL ENGINEERING

Copenhagen, Denmark

2016 - 2018

- Graduated with **"Highest Honours."**
- Master Thesis: "Hybrid motor imaginary brain computer interface and virtual reality based system for neurorehabilitation of stroke patients, employing deep learning classification" (Grade: 12/12), part of the thesis served as a basis for the publication [10].

## Budapest University of Technology and Economics

M.SC. MECHATRONICS ENGINEERING

Budapest, Hungary

2016 - 2020

- Graduated with **"Excellent with Highest Honors"**
- Master Thesis: "Motion Based Epileptic Seizure Classification with Deep Learning" (Grade: 5/5)

## Faculty of Engineering, University of Porto and INESC TEC

ERASMUS EXCHANGE SEMESTER

Porto, Portugal

Febr. - July 2019

## Budapest University of Technology and Economics

B.SC. MECHATRONICS ENGINEERING

Budapest, Hungary

2012 - 2016

- Graduated with **"Excellent"**
- Bachelor Thesis: "Analysis of amputated limb and prosthesis socket connection using the finite element method" (Grade: 5/5)

## Skills

|                              |                                                                      |
|------------------------------|----------------------------------------------------------------------|
| <b>Programming languages</b> | Python, Matlab, LaTeX, C/C++, C#                                     |
| <b>Software libraries</b>    | Pytorch, Tensorflow, Scikit-learn, Numpy, Matplotlib, OpenCV, Pandas |
| <b>Other</b>                 | Git, Docker                                                          |
| <b>Languages</b>             | English (C1), Hungarian (Native), German (Elementary)                |

## Publications

### JOURNAL ARTICLES

- [1] **T. Karácsny** et al. "Exploring Image and Skeleton-Based Action Recognition Approaches for Clinical In-Bed Classification of Simulated Epileptic Seizure Movements". In: *Expert Systems with Applications* (2025).
- [2] A. M. Loesch-Biffar, **T. Karácsny** et al. "Clinical application and new visualization techniques of 3D-quantitative motion analysis in epileptic seizures characterized by ictal automatic movements". In: *Epilepsy & Behavior* 170 (2025). † **Co-first author**.
- [3] **T. Karácsny** et al. "Deep learning methods for single camera based clinical in-bed movement action recognition". In: *Image and Vision Computing* 143 (2024).
- [4] **T. Karácsny** et al. "Novel 3D video action recognition deep learning approach for near real time epileptic seizure classification". In: *Scientific Reports* 12.1 (2022).

### PEER-REVIEWED CONFERENCE PROCEEDINGS

- [5] **T. Karácsny** et al. "NeuroKinect4K: A Novel 4K RGB-D-IR Video System with 3D Scene Reconstruction for Enhanced Epileptic Seizure Semiology Monitoring". In: *IEEE EMBC* (2024).
- [6] J. Carmona, **T. Karácsny** et al. "BlanketGen - A Synthetic Blanket Occlusion Augmentation Pipeline for Motion Capture Datasets". In: *IEEE ENBENG* (2023).
- [7] J. Carmona, **T. Karácsny** et al. "BlanketSet-A Clinical Real-World In-Bed Action Recognition and Qualitative Semi-Synchronised Motion Capture Dataset". In: *IEEE ENBENG* (2023).
- [8] **T. Karácsny** et al. "Deepepil: Towards an Epileptologist-Friendly AI Enabled Seizure Classification Cloud System based on Deep Learning Analysis of 3D videos". In: *IEEE BHI* (2021).
- [9] **T. Karácsny** et al. "A deep learning architecture for epileptic seizure classification based on object and action recognition". In: *IEEE ICASSP* (2020).
- [10] **T. Karácsny** et al. "Brain computer interface for neuro-rehabilitation with deep learning classification and virtual reality feedback". In: *Proceedings of the 10th Augmented Human International Conference* (2019).

### PRE-PRINTS AND SUBMISSIONS

- [11] **T. Karácsny** et al. "BlanketGen2-Fit3D: Synthetic Blanket Augmentation Towards Improving Real-World In-Bed Blanket Occluded Human Pose Estimation". In: *arXiv preprint arXiv:2501.12318* (2025).
- [12] R. Aslani, **T. Karácsny** et al. "Video-based Epileptic Seizure Classification: A Novel Multi-stage Approach Integrating Vision and Motion Transformer Deep Learning Models". In: *Submitted, Under review* (2025).

## Honors & Awards

### INTERNATIONAL AWARDS

- 2025 **Outstanding Reviewer - CVPR 2025, IEEE / CVF Computer Vision and Pattern Recognition Conference** *Nashville, TN, USA*  
Top ~5% of reviewers, as judged by Area Chairs, recognized for exceptional review quality and impact.
- 2024 **Innovation and Contribution to the field of AI** *Dubai, UAE*  
IEEE AI Research Hub Competition, at GITEX GLOBAL "Largest Tech & Startup Show in the World"

## Certificates

- 2024 **ACM Europe Summer School on HPC Computer Architectures for AI and Dedicated Applications**  
ACM, Association for Computing Machinery
- 2024 **From Portugal to the World Innovation-to-Business Journey**  
Cockrell School of Engineering, The University of Texas at Austin
- 2024 **INVICTA school of Vision, Computational intelligence, and patTern Analysis**  
INVICTA School
- 2021 **VISUM 2021 (online edition), the 9th edition of the Vision Understanding and Machine Intelligence summer school**  
VISUM summer school
- 2020 **VISUM 2020 - Summer School on Vision Understanding and Machine Intelligence**  
VISUM summer school

## Professional Services

### Reviewer

#### CONFERENCES

- CVPR 2025, ECCV 2024, MICCAI 2024-2025, IEEE ICASSP 2024–2025, IEEE AIVR 2020–2022, EAI Mobihealth 2022, IEEE TENCON 2019

#### JOURNALS

- Computer Methods and Programs in Biomedicine* (2023), *Neural Regeneration Research* (2023)

### Program Committee Member

#### WORKSHOP

- Workshop on AI and Advanced Human Sensing, AIMS Meeting 2021 (INESC TEC, FMUL, Virtual)

## Presentations/Invited talks

### Computer Vision in Human Motion Analysis for Biomedical Applications

BIOMEDICAL ENGINEERING SEMINARS, FEUP

*Porto, Portugal*  
2024

### 3D Scene Reconstruction and 4K RGB-D-IR Video For Epileptic Seizure Monitoring: Introducing NeuroKinect4K

CIÊNCIA 2024, ENCONTRO COM A CIÊNCIA E TECNOLOGIA EM PORTUGAL

*Porto, Portugal*  
2024

### In-bed action recognition for clinical diagnosis support: A two-stage, 3D motion capture and skeleton action recognition based approach

5TH DOCTORAL CONGRESS IN ENGINEERING

*Porto, Portugal*  
2023

- Best Paper award

### Explainable Deep Learning Based Epileptic Seizure Classification with Clinical 3D Motion Capture

CMU PORTUGAL

*Pittsburgh, USA*  
2023

- PhD Research Presentation During Visit of Elvira Fortunato, Portuguese Minister of Science, Technology and Higher Education

### Explainable Epileptic Seizure Classification: A 2-stage Pipeline

CMU PORTUGAL SUMMIT

*Lisbon, Portugal*  
2022

### 3D Motion capture technologies for clinical patient monitoring – a short summary

CIÊNCIA 2022, ENCONTRO COM A CIÊNCIA E TECNOLOGIA EM PORTUGAL

*Lisbon, Portugal*  
2022

### An Epileptologist-Friendly Cloud-Based Remote 3Dvideo-EEG Processing Environment For Quantified Semiology Analysis

JUNTOS PELO FERNANDO CONFERENCE

*Lisbon, Portugal*  
2020

## Volunteer work

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### **INESC TEC - Institute for Systems and Computer Engineering, Technology and Science**

*Porto, Portugal*

PHD STUDENT REPRESENTATIVE

*Nov. 2024 - Present*

- Building PhD student representation with 11 representatives from 11 centers, advocating for student needs, and bridging communication with the INESC TEC board.

### **Collage of Engineering - Student Society, Budapest University of Technology and Economics**

*Budapest, Hungary*

VICE-PRESIDENT

*2015 - 2016*

- Led a student society of ≈400 members, oversaw and coordinated it with 13 section leaders, negotiated and secured industry partnerships, and organized large-scale events.

### **Section of Mechatronics - Student Society, Budapest University of Technology and Economics**

*Budapest, Hungary*

MEMBER, TEAM LEADER, VICE PRESIDENT

*2012 - 2016*

- Managed section activities, mentored team members, and built partnerships with external stakeholders.